THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The pair $kx + 7y = k - 7$ and $28x + ky = k$ has infinitely many solutions when $k =$ 1

(a) -14 **(b)** 14 **(c)** 7 **(d)** 28

2. If the roots of $x^2 - 12x + k = 0$ differ by 4, then $k =$ 1

(a) 128 **(b)** 64 **(c)** 32 **(d)** 40

3. The largest number dividing 75, 123 and 195 leaving remainder 3 in each case is: 1

(a) 6 **(b)** 48 **(c)** 24 **(d)** 12

4. If α, β are zeroes of $x^2 - 8x + 7$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 56 **(b)** 49 **(c)** 15 **(d)** 64

5. If $13x + 5y = 67$ and $6x + 7y = 45$, then $x + y =$ 1

(a) 7 **(b)** 4 **(c)** 3 **(d)** 12

6. For which value of k does $x^2 + kx + 16 = 0$ have equal roots (take $k > 0$)? 1

(a) 8 **(b)** 32 **(c)** 16 **(d)** 4

7. The smallest positive integer by which 18 must be multiplied to obtain a perfect square is: 1

(a) 3 **(b)** 6 **(c)** 18 **(d)** 2

8. If α, β are zeroes of $x^2 - 9x + 8$, then $|\alpha - \beta|$ is: 1

(a) 8 **(b)** 14 **(c)** 9 **(d)** 7

9. The area (in sq. units) of the triangle formed by the line $3x + 4y = 12$ and the coordinate axes is: 1

(a) 12 **(b)** 2 **(c)** 72 **(d)** 6

10. If $x = 2$ is a root of $kx^2 - 3x - 2 = 0$, then $k =$ 1

(a) -2 **(b)** 2 **(c)** $\frac{1}{2}$ **(d)** 1

11. Which of these is a rational number? 1

(a) $\sqrt{5} + \sqrt{6}$ **(b)** $\sqrt{7}$ **(c)** $5\sqrt{8}$ **(d)** $\frac{\sqrt{75}}{\sqrt{3}}$

12. If α, β are zeroes of $x^2 - 7x + 6$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) $\frac{49}{6}$ **(b)** $\frac{7}{6}$ **(c)** $\frac{37}{6}$ **(d)** 37

13. If $x + y = 9$ and $xy = 20$, then $x^2 + y^2 =$ 1

(a) 41 **(b)** 81 **(c)** 61 **(d)** 121

14. The equation $kx^2 + 4x + 2 = 0$ has real roots when: 1

(a) $k > 0$ **(b)** $k < 2$ only

(c) $k \leq 2$ and $k \neq 0$ **(d)** $k \geq 2$

15. A number ends with exactly 3 zeros if its prime factorisation (other factor 3) is: 1

(a) $3^3 \times 7$ **(b)** $2^4 \times 5^2 \times 3$ **(c)** $2^4 \times 5^3 \times 3$ **(d)** $2^2 \times 5^3 \times 3$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the points where the line $3x + 2y = 12$ meets the coordinate axes and the area of the triangle so formed. 2

17. The sum of a number and its reciprocal is $\frac{5}{2}$. Find the number. 2

18. Two numbers have HCF 15 and are in the ratio $4 : 7$. Find the numbers and their LCM. 2

19. If α, β are zeroes of $x^2 - 5x + 6$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

20. Find k so that the pair $kx + 3y = k - 3$, $12x + ky = k$ has no solution. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Solve: $\frac{3}{x} + \frac{4}{y} = \frac{11}{4}$ and $\frac{5}{x} + \frac{2}{y} = \frac{9}{4}$. 3

22. Solve for x: $\frac{x+2}{x-2} + \frac{x-2}{x+2} = \frac{26}{5}$. 3

23. Find the greatest number that divides 1157 and 1397 leaving remainder 5 in each case. 3

24. The zeroes of $x^2 - 12x + c$ are α and β with $\alpha - \beta = 6$. Find c. 3

OR

Find a and b for which the pair $5x + 3y = 4$ and $(a+b)x + (a-b)y = 12$ has infinitely many solutions. 3

25. For what values of k does $x^2 + 7x + k = 0$ have real roots? Find the greatest integer value of k. 3

OR

Is $6 \times 7 \times 8 + 8$ a composite number? Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Three sets of English, Hindi and Mathematics books have 144, 192 and 288 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

The zeroes of $x^3 - 9x^2 + 23x - 15$ are in arithmetic progression. Find them. 5

27. A boat goes 21 km upstream and 33 km downstream in 10 hours, and 12 km upstream and 55 km downstream in 9 hours. Find the speed of the boat in still water and the speed of the stream. **5**

OR

Two taps running together fill a tank in $\frac{24}{5}$ hours. The larger tap alone takes 4 hours less than the smaller one. Find the time each takes alone. **5**

END OF QUESTION PAPER **Total: 50 Marks**

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